Structural Correspondence Between Order Formation in Physics and the Five-Stage Model

2026-03-23

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1. Purpose and Question of the Investigation

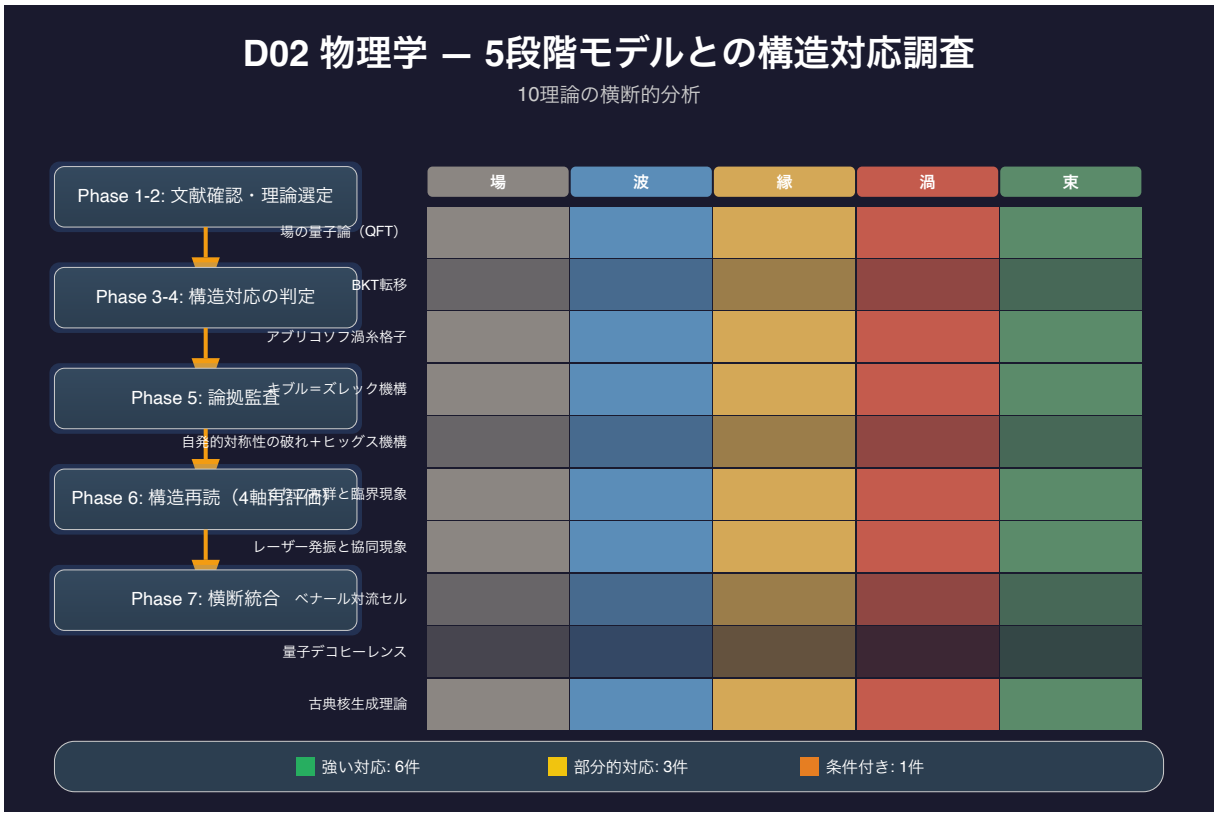

Figure 1: Structural Correspondence Between Order Formation in Physics and the Five-Stage Model —

This report was prepared to answer the following question.

> Do theories in physics structurally correspond to the five-stage model (Field ( ) → Wave ( ) → Edge ( ) → Vortex ( ) → Bundle ( ))?

The five-stage model is a framework that describes the structure of creative processes in terms of five stages: Field, Wave, Edge, Vortex, and Bundle. In this investigation, physics, especially theories related to phase transitions, critical phenomena, and self-organization, was examined in order to consider its structural similarities to this model.

What this report addresses: Whether theories in physics structurally correspond to the five-stage model, how strong that correspondence is, and what implications can be drawn from it.

What this report does not address: Textbook-style explanations of physics, evaluation of whether individual theories are correct, or recommendations for specific applications.

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2. Method of Investigation

Scope and Selection Criteria

Within physics, we selected ten theories and phenomena, focusing on theories that describe how order comes into being. Specifically, the ten cases are quantum field theory (QFT), the

BKT transition, the Abrikosov vortex lattice, the Kibble-Zurek mechanism (KZM), spontaneous symmetry breaking (SSB) and the Higgs mechanism, the renormalization group and critical phenomena, laser oscillation and cooperative phenomena, Benard convection cells, quantum decoherence, and classical nucleation theory.

Evaluation Method

For each theory, we examined whether there exists a physical process corresponding to each stage of the five-stage model: Field, Wave, Edge, Vortex, and Bundle. These determinations were made not from a single viewpoint, but from multiple independent perspectives, including similarity of theoretical structure, the temporal ordering of processes, and the role of boundary conditions. Conclusions were then formed through a comparative assessment that integrated these distinct lines of analysis.

Criteria for Assessment

- **Strong structural correspondence**: A physical process corresponding to all five stages can be identified, and the mechanisms of transition between stages also show correspondence.
- **Partial correspondence**: Some stages have clear counterparts, but correspondence in other stages is weak or absent.
- **Conditional correspondence**: Correspondence may be established depending on the interpretation, but alternative interpretations are also possible.

Limits of the Investigation

This investigation does not cover the whole of physics. It is biased toward particle physics, condensed matter physics, and nonequilibrium physics, while astrophysics and many areas of fluid dynamics are outside its scope. Moreover, even if structural correspondence is confirmed, this investigation alone cannot determine whether it is an accidental similarity or a reflection of a deeper shared structure.

In addition, quantum field theory (QFT) is the source of inspiration for the five-stage model. Its close correspondence with the model is therefore a consequence of that origin and cannot be regarded as independent evidence. To avoid the risk of circular reasoning, this project tests whether the same structure can be independently confirmed across 29 other domains.

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3. Overview of the Model: What Is the Five-Stage Model?

The five-stage model describes the structure of creative processes in terms of the following five stages.

Field - An undifferentiated state. Direction and structure are not yet determined, and all possibilities coexist.

Wave - A stage in which multiple directions diverge and compete. Differences arise within the field, and exploration begins.

Edge - A state of tension in which opposing elements coexist without converging on either side. Its characteristic feature is remaining in unresolved contradiction, that is, avoiding premature resolution and allowing the opposition to persist.

Vortex - A stage in which a new coherence (order) spontaneously arises from within that tension. Rather than having order imposed from outside, order emerges from the interactions among elements.

Bundle - A stage in which form becomes determinate and stabilizes as a reusable structure. The result of creation solidifies and takes on a form that can be communicated externally.

This model is not limited to a particular scale, such as individual thought, organizational decision-making, or physical processes, but is proposed as a general structural description of order formation.

Some readers may feel discomfort with these definitions themselves as they proceed through this report. That discomfort is an important perspective in considering the validity of the model. One possible way of reading what follows is to continue while retaining that discomfort.

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4. Results of the Investigation: Overall Picture

Figure 2: Structural Correspondence Between Order Formation in Physics and the Five-Stage Model — ×5

Overview

Evaluation of the ten theories and phenomena found strong structural correspondence in nine cases and conditional correspondence in one case. One additional item, Pauli matrices and symmetry breaking, was excluded from evaluation because it overlaps with the mathematics domain (D01).

Theory / phenomenon	Assessment	Edge correspondence	Notes
Quantum field theory (QFT)	Strong structural correspondence	Strong	Source of inspiration for the five-stage model
BKT transition	Strong structural correspondence	Strong	Topological phase transition
Abrikosov vortex lattice	Strong structural correspondence	Strong	Vortex → Bundle can be directly observed
Kibble-Zurek mechanism (KZM)	Strong structural correspondence	Strong	Necessary structure grounded in causality
SSB + Higgs mechanism	Strong structural correspondence	Moderate	The description of indeterminacy at Edge is somewhat weak
Renormalization group and critical phenomena	Strong structural correspondence	Strong	Describes the process itself
Laser oscillation	Strong structural correspondence	Strong	Threshold can be described quantitatively
Benard convection cells	Strong structural correspondence	Moderate	Raises the issue of multistability
Quantum decoherence	Conditional correspondence	Moderate	The Vortex stage is absent
Classical nucleation theory	Strong structural correspondence	Strong	The interface exists as a physical reality

Overall Tendency

In the physics domain, mechanisms centered on critical phenomena and phase transitions, namely mechanisms in which new order emerges as a system passes through a critical point or interface, are overwhelmingly dominant: 9 of the 10 cases belong to this type. This is consistent with the nature of physics as a discipline, one that describes in experimentally traceable form how order comes into being. At the same time, it is necessary to note whether this concentration reflects the essence of physics itself or a bias in theory selection.

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5. Results of the Investigation: Findings by Theory

5-1. Quantum Field Theory (QFT): The Privileged Position of a Source of Inspiration

Quantum field theory is a theory that takes quantum fields as the fundamental entities and describes particles as excited states of fields. The vacuum is understood not as emptiness, but

as a filled field containing quantum fluctuations. Interactions are characterized by coupling constants and are visualized as vertices in Feynman diagrams within perturbative expansions.

Its correspondence with the five-stage model can be organized as follows. The quantum field itself, namely the vacuum as a superposition of all modes, corresponds to Field. Vacuum fluctuations, such as the creation and annihilation of virtual particle pairs, correspond to Wave, and interactions between fields, namely couplings, correspond to Edge. The fact that, without interaction, nothing emerges beyond free fields agrees with the model's structure in which, without Edge, neither Vortex nor Bundle arises. The formation of resonant states or bound states, for example color confinement in QCD, corresponds to Vortex, while stable particles and spectra correspond to Bundle.

At the same time, because QFT is the source of inspiration for the five-stage model, the close fit is a consequence of that origin. Its value as independent evidence depends on the results of examination across the other 29 domains.

One QFT-specific insight deserving attention is the path-integral formalism's presentation of all possibilities being summed in parallel. Unlike the temporal sequentiality suggested by the five-stage model, QFT allows all paths to contribute simultaneously, with a particular result selected through interference. This difference suggests that the five stages should be understood not as a temporal sequence, but as structural layers.

5-2. Abrikosov Vortex Lattice: Direct Observation from Vortex to Bundle

When a magnetic field is applied to a type II superconductor, the magnetic field penetrates the superconductor in the form of filamentary vortex lines. Around each vortex line, the phase changes by 2 over one full circuit, and each line carries the discrete unit known as the magnetic flux quantum (hc/2e). The vortex lines interact with one another and spontaneously form a triangular lattice. This triangular lattice can be directly observed by electron microscopy, and Abrikosov received the 2003 Nobel Prize in Physics for this contribution (Abrikosov 1957, observation by Essmann-Trauble 1967).

The correspondence with the five stages is clear. The order-parameter field and the magnetic field are the Field stage, and local destabilization is the Wave stage. Phase winding, namely a topological relational constraint, together with interactions among vortex lines, corresponds to Edge. Quantized vortex lines correspond to Vortex, and the triangular lattice of vortex lines corresponds to Bundle.

The distinctive value of this case lies in the fact that the process from Vortex to Bundle is directly visualized as the geometric order of a triangular lattice. It is a concrete example showing that Bundle is not merely the determination of a direction, but an order with a spatial arrangement. At the same time, the discreteness imposed by magnetic flux quantization, namely that vortices appear not continuously but in quantized units, is an aspect not found in the five-stage model and provides a clue to the model's limits of applicability.

5-3. Kibble-Zurek Mechanism (KZM): The Physical Necessity of the Stages

The Kibble-Zurek mechanism is a theory stating that, when a phase transition is crossed at a finite rate, spatially uniform order cannot be achieved because of the constraints of causality, and domain boundaries or topological defects therefore form inevitably. Critical slowing down near the critical point, where the response speed of the system, namely the relaxation time, diverges, determines the domain size, and this in turn determines the density of defects. This mechanism has also been confirmed experimentally using trapped-ion crystals (Ulm et al. 2013). It was originally proposed by Kibble (1976) in the context of cosmology and by Zurek (1985) in the

context of condensed matter, and it is a candidate for the broadest scale-crossing range among all 30 domains, extending from cosmology to tabletop experiments.

The Field stage corresponds to the high-symmetry phase, that is, the state before the transition. The Wave stage corresponds to critical fluctuations. The Edge stage corresponds to domain boundaries, namely contact regions in which the phases of adjacent domains fail to align. The Vortex stage corresponds to the formation of topological defects, and the Bundle stage corresponds to defect networks and their scaling laws.

KZM provides the most important insight in the physics domain because it offers a physical basis for why the stages must be traversed. The claim that spatial nonuniformity is unavoidable when a phase transition is crossed at a finite rate because of causality may serve as a physical argument for the necessity of staged processes. However, whether physical phase transitions and creative processes can legitimately be identified remains untested, and the claim does not go beyond the level of hypothesis.

5-4. Renormalization Group and Critical Phenomena: A Description of the Process Itself

The renormalization group is a framework for systematically describing the relations among scales in a physical system. By coarse-graining, that is, averaging over short-distance degrees of freedom, one derives an effective theory, and by repeating this operation one analyzes structure across scales. Beginning with Kadanoff's (1966) block-spin transformation, the framework was systematized by Wilson (1971), who received the Nobel Prize in Physics in 1982. At the critical point, the correlation length diverges to infinity and all scales become coupled. The universality of fixed points and critical exponents, namely the classification of phenomena that display the same critical behavior regardless of microscopic detail, is a central achievement of this theory.

The correspondence with the five stages is as follows. A system near criticality corresponds to Field. Fluctuations differentiated by scale correspond to Wave. Renormalization-group flow, namely the operation that rewrites relations among scales, corresponds to Edge. Scaling near fixed points corresponds to Vortex, and universality classes, namely the classification of macroscopic behavior independent of microscopic detail, correspond to Bundle.

An important insight specific to the renormalization group is its threefold classification of operators. Under the renormalization group, operators are classified as relevant, irrelevant, or marginal. A marginal operator in particular has the property that its behavior cannot be determined at the stage of linear analysis; only after higher-order corrections are calculated does its behavior become settled. This structure, in which the result is not decided at first pass and requires additional analysis, structurally corresponds to the characteristic of Edge as a stage that avoids premature judgment and leaves opposition unresolved.

Moreover, because the renormalization group is a theory that describes not a state but the process itself, it is also a rare theory that can capture operationally what is happening at Edge.

5-5. Classical Nucleation Theory: Edge as a Physically Real Interface

Classical nucleation theory describes the transition from a metastable state, such as a supercooled liquid or supersaturated solution, to a stable phase through the formation of a critical nucleus. The critical nucleus is characterized by the critical size $r^* = 2 / |\Delta Gv|$, at which the gain in bulk free energy and the cost of interfacial energy are in balance (is the interfacial tension and ΔGv is the bulk free-energy difference). Nuclei smaller than the critical nucleus disappear, while larger nuclei grow irreversibly. This theory was formulated by Becker-Doring (1935) and has broad applications, from crystallization, cloud formation, and beer-foam formation to industrial processes.

Its correspondence with the five stages is intuitive. The metastable phase is the Field stage. Thermal fluctuations in density or composition, namely the repeated generation and disappearance of microscopic nuclei, are the Wave stage. The critical nucleus and its interface, namely the region where the old and new phases are in physical contact and the branching between growth and extinction occurs, correspond to Edge. Irreversible growth after the critical size is exceeded corresponds to Vortex, and the establishment of the stable phase, such as a crystal grain, droplet, or bubble, corresponds to Bundle.

The distinctive value of this case lies in the fact that Edge appears not as a conceptual metaphor, but as a spatially real physical interface. The interface of the critical nucleus has a thickness of several molecular layers, and within it the old and new phases coexist. This is a physical example showing that Edge is not a line but a region. Whereas Edge in other cases in physics, such as renormalization-group flow, mode competition, and interaction terms, is a conceptual structure, the Edge of nucleation is a spatial interface that can be physically defined.

5-6. Other Theories

The remaining four theories also display structural correspondence in outline.

BKT transition: This is a topological phase transition in a two-dimensional system. On the low-temperature side, vortex pairs are bound, and at the transition temperature those pairs unbind, producing a new phase, the free-vortex plasma. The binding of vortex pairs corresponds to Edge, and the indeterminacy of whether that binding will be released embodies the characteristic of Edge. At first glance, the free-vortex plasma corresponding to the Bundle stage appears disordered, but it is a new phase topologically distinct from the low-temperature phase and therefore signifies the establishment of a new order (Berezinskii 1971, Kosterlitz-Thouless 1973).

SSB + Higgs mechanism: This is the process in which, from a ground state with high symmetry, one vacuum is spontaneously selected through spontaneous symmetry breaking, Goldstone modes appear, and gauge fields acquire mass through the Higgs mechanism. Goldstone modes, which appear as traces left by broken symmetry and couple to other fields, possess Edge-like qualities as records that symmetry is not simply "broken" but "hidden." However, the description of indeterminacy is somewhat weak, and the correspondence is therefore moderate (Goldstone 1961, Higgs 1964).

Laser oscillation: Energy is injected into an ensemble of atoms by pumping, a population inversion is formed, and once the threshold is exceeded a cascade of stimulated emission generates coherent light. Haken (1983) formulated this process as an analogy to phase transition, within synergetics. Mode competition near the threshold, namely a state in which multiple modes coexist and it has not yet been determined which one will dominate, corresponds to Edge. The process in which the dominant mode cooperatively drives the other atoms, known as the slaving principle, corresponds to Vortex, and coherent light corresponds to Bundle. The distinctive feature of the laser case is that the conversion from Edge to Vortex can be described quantitatively, in terms of the critical pump strength.

Benard convection cells: When a fluid layer heated from below exceeds the critical Rayleigh number, it spontaneously forms convection cells (Rayleigh 1916, Benard 1900). Patterns such as hexagons, rolls, and spirals appear as instances of symmetry breaking. This is the most classical example of a dissipative structure, and the process disorder → criticality → self-organization → stable pattern is an intuitive visualization of the five-stage model. Competition among pattern modes near criticality is Edge-like, but the description of a network of relations is somewhat weak, so the correspondence is assessed as moderate.

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6. Cross-Cutting Patterns

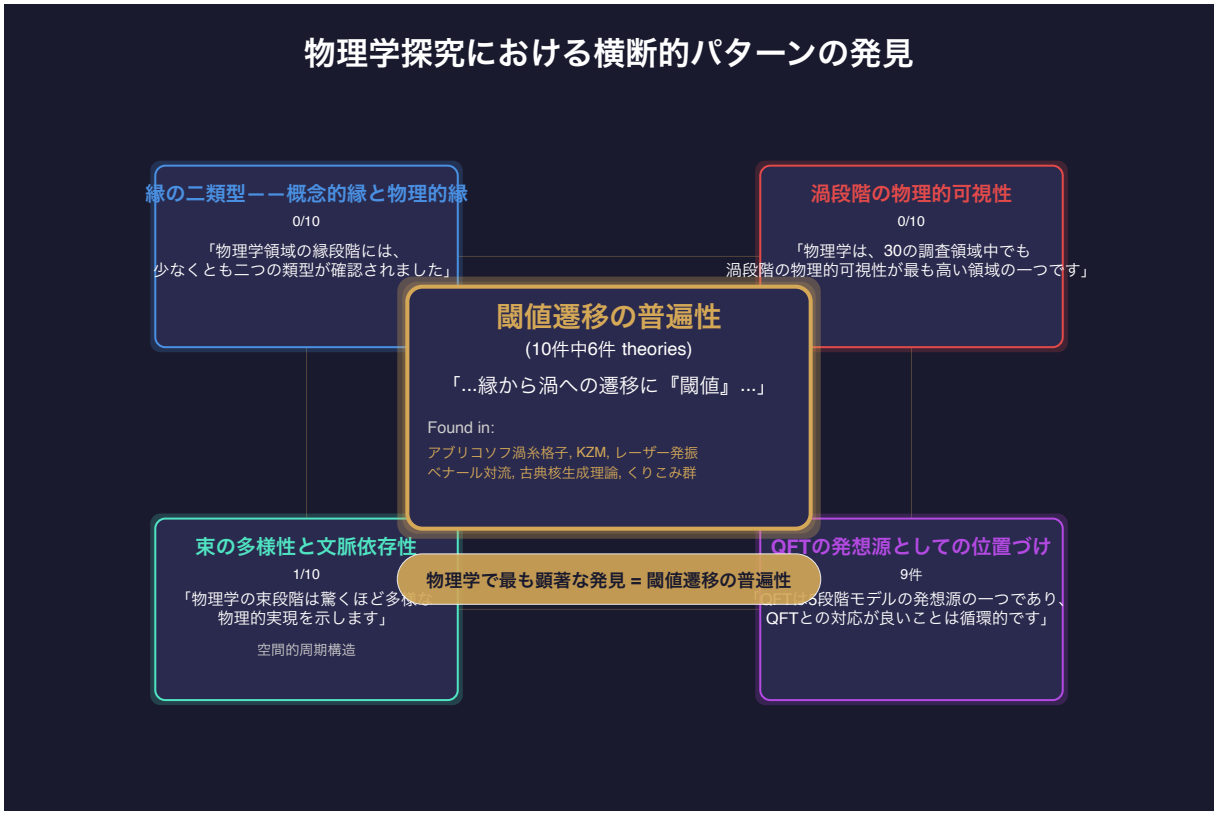

Figure 3: Structural Correspondence Between Order Formation in Physics and the Five-Stage Model —

When the ten physical theories are examined across the domain, the following patterns repeatedly appear.

6-1. The Universal Role of Critical Points, Interfaces, and Thresholds

In 9 of the 10 cases, some kind of critical point, interface, or threshold was identified as the physical process corresponding to the Edge stage. Bound vortex pairs in the BKT transition, domain boundaries in KZM, mode competition near the threshold in lasers, and the critical nucleus in nucleation theory all function as regions between two states in which neither side is decisively fixed. The fact that physics so consistently describes such phenomena in terms of criticality suggests that Edge in the five-stage model is deeply connected to the structure of physical critical points.

6-2. The Spontaneity of the Vortex Stage

One feature repeatedly confirmed across multiple theories is that the process corresponding to the Vortex stage is not imposed from outside, but arises spontaneously from within the system. The cascade of stimulated emission in laser oscillation, the unbinding of vortices in the BKT transition, and the formation of vortex lines in the Abrikosov lattice all exhibit order emerging from interactions among the elements of the system without any external command. This spontaneity agrees with the definition of the Vortex stage in the five-stage model, namely that a new coherence spontaneously arises.

6-3. The Diversity of the Bundle Stage

The physical entities corresponding to the Bundle stage differ greatly from theory to theory: coherent light in the laser, the triangular lattice in the vortex lattice, universality classes in the renormalization group, stable particles in QFT, convection patterns in Benard convection, and crystal grains or droplets in nucleation. These all share the commonality of being stable structures, but their concrete forms are extremely diverse. This confirms that the Bundle stage is not a single fixed form, but takes different forms depending on context.

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7. Unresolved Questions

7-1. A Route That Does Not Pass Through the Vortex Stage

In quantum decoherence, one can observe the path Field, namely quantum superposition, → Wave, namely the beginning of interaction with the environment, → Edge, namely the selection of pointer states, that is, the process by which states stable against the environment are singled out, → Bundle, namely the classically definite state. However, no clear process corresponding to the Vortex stage could be identified. It is possible to interpret quantum Darwinism, namely the proliferative diffusion of information into the environment, as corresponding to the Vortex stage, but this has not yet been verified. This case raises the question of whether the five stages must always all be traversed, or whether routes that skip stages may exist.

7-2. Multistability and the Uniqueness of Bundle

In Benard convection cells, it is known that after the critical Rayleigh number is exceeded, multiple patterns such as hexagons, rolls, and spirals can coexist stably. If the Bundle stage means that form becomes determinate, then a situation in which multiple patterns exist stably at the same time calls the uniqueness of Bundle into question.

7-3. Circular Reasoning with the Source of Inspiration

QFT is the source of inspiration for the five-stage model, and the close correspondence between the two is therefore a consequence of that origin. The value of this investigation does not lie in the correspondence with QFT itself, but in whether the same structure can be found independently in the other 29 domains. If a structure extracted from QFT is repeatedly confirmed even in domains unrelated to that source of inspiration, then that structure is more likely to be not something specific to physics, but a more general pattern of order formation. This verification is built into the design of the project as a whole.

7-4. Structural Correspondence or Similarity of Labels?

This investigation alone cannot determine whether critical phenomena in physics and Edge in the five-stage model are structurally identical, or whether the labels “critical point” and “Edge” merely resemble one another. To answer this question, it is necessary to examine whether the same structure can be confirmed in domains other than physics, and if so, how precisely the structures match.

7-5. The Structural Difference Between First-Order and Continuous Phase Transitions

The addition of classical nucleation theory, a first-order phase transition, raises the question of how the structural difference from continuous phase transitions, namely the process of crossing an energy barrier in the first-order case and continuous change driven by critical fluctuations in

the continuous case, affects correspondence with the five stages. It is possible that the nature of the Edge stage differs between the two, but this has not yet been examined.

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8. Conclusion

The investigation in the physics domain confirms the following.

Of the ten theories and phenomena, nine showed strong structural correspondence with the five-stage model. In particular, theories related to critical phenomena and phase transitions possess physical processes corresponding to each stage of Field → Wave → Edge → Vortex → Bundle, and correspondence is also visible in the transition mechanisms between stages.

The degree of certainty differs by stage. In many theories, the three stages of Field, Wave, and Bundle have clear counterparts and can be reported as determinate findings. The Edge stage is also confirmed in every theory that showed strong structural correspondence, in the form of critical points, interfaces, or thresholds, and this too can be reported as a strong tendency. The Vortex stage can be identified in many theories as spontaneous order formation, but one case, quantum decoherence, was also found in which the Vortex stage is absent. Accordingly, the necessity of the Vortex stage remains at the level of hypothesis.

Physics is a particularly important domain for the five-stage model for three reasons. First, correspondence can be traced in both theory and experiment, as in electron-microscopic observation of the Abrikosov vortex lattice and trapped-ion experiments on KZM. Second, the same structure recurs in phenomena at different scales, from elementary particles to cosmology. Third, physics may provide a physical argument for the necessity of stages, through KZM's account of the inevitability of staged formation under causality.

At the same time, three reservations should always be kept in mind when interpreting this conclusion. The fact that QFT is the source of inspiration means that the strong correspondence in the physics domain must be discounted to some degree. The bias of theory selection toward critical phenomena means that the conclusion may not represent physics as a whole. And the validity of identifying physical processes with creative processes cannot be tested within the scope of this investigation.

To state the degree of confidence in this report honestly: "There exists structural correspondence between theories of order formation in physics and the five-stage model" can be reported as a determinate finding based on repeated confirmation across multiple independent theories. "This correspondence reflects a deeper shared structure that goes beyond accidental similarity" cannot be judged from the investigation of the physics domain alone and requires cross-domain examination with the other 29 domains.